High-Performance C++ Dual-Pivot Quicksort

A Drop-in Replacement for `std::sort`

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Final Year Project — Final Presentation

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Agenda

1. **The Product** — what `dpqs::sort` is and what it delivers
2. **Research Gap & Methodology** — why another sort, how we measured
3. **Sequential Walkthrough** — seven guards, in execution order
   - Early termination · type dispatch
   - Insertion leaves · run-merging · heapsort fallback
   - Dual-pivot · Dutch National Flag
4. **Parallel Path** — work-stealing, scaling curve
5. **VTune Root-Cause** — why 16T regresses
6. **Engineering, Limitations, Conclusion**
7. **Q & A**

The Product: `dpqs::sort(...)`

What it is

- Header-only C++17 sorting library
- Drop-in replacement for `std::sort`
- Same iterator + comparator contract

What it delivers

- = parity on random 32-bit `int`
- ∼**10**× on reverse-sorted (near-linear)
- **4.72**× peak parallel speedup
- Zero dependencies — one `#include`

```cpp
#include "dual_pivot_quicksort.hpp"

std::vector<int> v = {...};

// Auto-parallel (default)
dual_pivot::sort(v);

// Iterator form
dual_pivot::sort(v.begin(), v.end())

// Force sequential
dual_pivot::sort(v, 1);

// Custom comparator
dual_pivot::sort(v, std::greater<int
```

Research Gap — Why Another Sort?

The Java–C++ discrepancy

- Java adopted Yaroslavskiy's dual-pivot quicksort in **JDK 7 (2011)** for primitive types
- C++ `std::sort` (libstdc++, libc++, MSVC) still uses **single-pivot Introsort** [3]
- No widely adopted, STL-compliant dual-pivot implementation exists in production C++

Does the Java win translate to C++?

- Java: expensive comparisons → "more comparisons, fewer memory accesses" is a clear win
- C++: templates inline comparators → comparisons are near-free
- **Open question:** does the memory-traffic reduction survive?

What this project does

1. **Engineer** a modern, STL-compliant C++17 dual-pivot sort
2. **Measure** whether it can beat `std::sort` in a native environment
3. **Extend** it with adaptive run-merging, DNF, and work-stealing parallelism

Existing C++ DPQS code is non-generic, experimental, or pre-C++11 — not production-grade.

Methodology

Hardware

- Intel Core i5-12600KF
- 6 P-cores (SMT) + 4 E-cores — **10 physical / 16 logical**
- L1-D: 6 × 48 KB (P) + 4 × 32 KB (E)
- L1-I: 6 × 32 KB (P) + 4 × 64 KB (E)
- L2: 6 × 1.25 MB (P) + 2 × 2 MB (E-cluster)
- **L3: 20 MB** shared
- 32 GB DDR5 @ 2992.7 MHz

Compiler

- GCC 13.3.0 (MinGW-w64)
- `-std=c++17 -O2 -march=native -DNDEBUG`
- `-pthread`

Workload — 7 872 configurations

- **6 algorithms** — `std::sort`, `dual_pivot_sequential`, `dual_pivot_parallel_{2,4,8,16}`
- **4 data types** — `int`, `int8_t`, `int16_t`, `double`
- **8 patterns** — random, nearly-sorted, reverse-sorted, organ-pipe, sawtooth, many-duplicates {10, 50, 90}%
- **41 sizes** — log-spaced $10^3 \rightarrow 10^7$

Randomness

- 10 seeds (42…51)
- **3 warmups + 10 timed** per seed
- Representative = **median of per-seed minima** (100 samples)

Structured patterns

- 30 iterations, min-time

Section 3 — Sequential Walkthrough

1. The Product
2. Research Gap & Methodology
3. **Sequential Walkthrough**
4. Parallel Path
5. VTune Root-Cause
6. Engineering, Limitations, Conclusion
7. Q & A

Execution Flow — What Happens Inside `dpqs::sort(...)` [1]

```
sort(container | iter,iter | ptr,len)
        |
        v
[Step 0]  Normalize API overloads
          -> sort(T* a, parallelism, lo, hi, Compare)
        |
        v
[Step 1]  Guards: null / range / early-termination scan  --> DONE (sorted)
        |
        v
[Step 2]  Type dispatch (if constexpr)
        |--- int8/int16 ---------> counting_sort()        --> DONE
        |--- float/double -------> sort_floats()
        |--- parallel eligible --> parallelQuickSort()    --> §3
        |--- otherwise ----------> sort_sequential()
```

Steps 3–5 continue on the next slide.

Execution Flow (cont.) — Steps 3, 4, 5

```
[Step 3]  Compute max_depth = 2 * log2(n) * DELTA
          (introsort safety net)
        |
        v
[Step 4]  Recursive core — guards 4.1 ... 4.7
          (walked in detail on next slides)
        |
        v
[Step 5]  Return (sorted in place)
```

We walk the numbered steps in execution order, and show the benchmark number each step earns.

Step 1 — Early Termination (Free O(n) Win)

Guard: single linear scan before any sorting work

- `low >= high` → return
- Null check / range validation
- **`checkEarlyTermination`:** O(n) scan; if already sorted → done

Result — fully-sorted input, 10^7 int

`std::sort`	≈ baseline (still partitions)
`dpqs`	**near-instant (O(n))**

Takeaway
The cheapest guard wins the easiest case outright.

"Data is already sorted" is more common in production than people think — log ingestion, prefix arrays, time-series append.

Step 2 — Type Dispatch (Compile-Time `if constexpr`)

Type	Strategy	Complexity
`int8_t`, `int16_t`	**Counting sort** [6]	O(n + k)
`float`, `double`	`sort_floats` (NaN / ±0 aware)	O(n log n)
Parallel-eligible	`parallelQuickSort` (§3)	O(n log n / p)
Otherwise	Dual-pivot quicksort [2]	O(n log n)

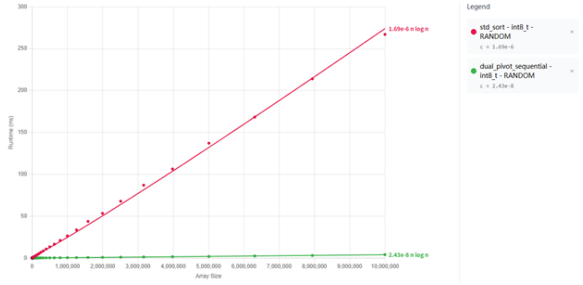


Figure 1: int8_t — dpqs vs std::sort

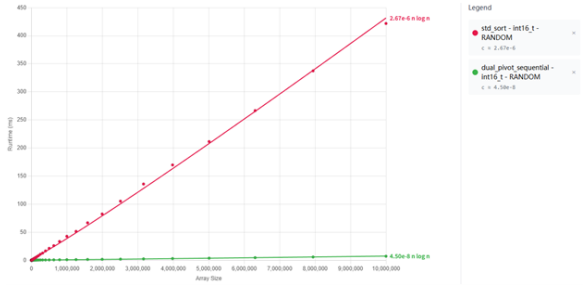


Figure 2: int16_t — dpqs vs std::sort

Step 4 — Recursive Core: Seven Guards, in Order

```
sort_sequential(lo, hi, leftmost):
    size = hi - lo
    if size < 65  and !leftmost          -> 4.1  mixed (pin) insertion sort
    if size < 45  and leftmost           -> 4.2  plain insertion sort
    if size > MIN_TRY_MERGE_SIZE:
        if try_merge_runs(...)           -> 4.3  Timsort-style run merge  <DONE>
    if depth >= max_depth                -> 4.4  heapsort fallback [3]    <DONE>

    # --- pivot machinery ---
    sort5_network(a[e1..e5])             -> 4.5  9-comparator network
    if a[e1] < a[e2] < a[e3] < a[e4] < a[e5]:
        dual_pivot_partition(P1=e1, P2=e5)   -> 4.6a  3 regions
    else:
        dnf_partition(P=e3)                  -> 4.6b  [<P][=P][>P]
    # --- recurse (4.7): loop on largest, recurse on two smaller ---
```

Each guard is shown next with the input pattern where it dominates.

4.1 / 4.2 — Insertion-Sort Leaves (Classical)

What & why

- `size < 65` (non-leftmost) → **mixed/pin insertion**; left pivot is a sentinel so no bound check
- `size < 45` (leftmost) → **plain insertion**
- Every leaf of the quicksort tree hits one of these

Result — random int, 10^7

Algorithm	Speedup
`std::sort`	baseline
`dpqs`	$\approx$ **1.1**$\times$

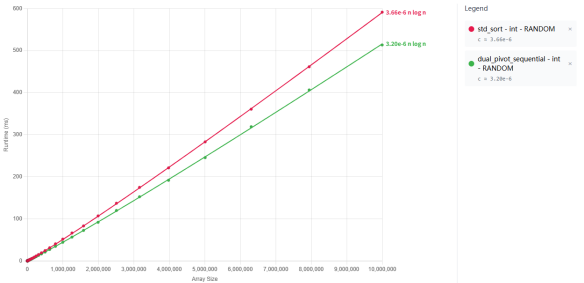


Figure 3: Random int32 — dpqs vs std::sort

4.3 — Adaptive Run-Merging (Timsort Lineage [4])

Scan for ascending / descending / constant runs

- If first run < `MIN_FIRST_RUN_SIZE` → **bail out fast** (random data pays tiny cost)
- Else merge runs in $O(n \log r)$ — **no partitioning, no recursion spawned**

r = *number of monotone runs detected in the input* ($1 \leq r \leq n/2$). *Nearly-sorted* ⇒ r *small* ⇒ *near-linear.*

Pattern	Speedup
Nearly-sorted	$\approx 0.74\times$ (slower)
Reverse-sorted	$\approx 10\times$ (near-linear)
Organ-pipe / Sawtooth	see figures

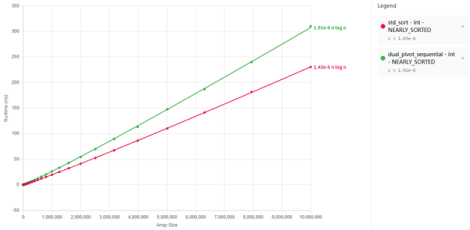


Figure 4: Nearly-sorted

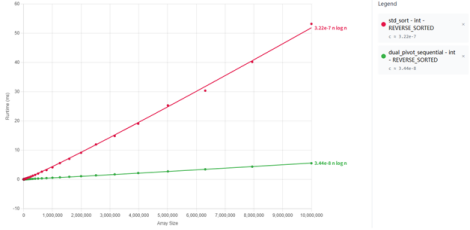


Figure 5: Reverse-sorted

4.3 (cont.) — Organ-Pipe & Sawtooth

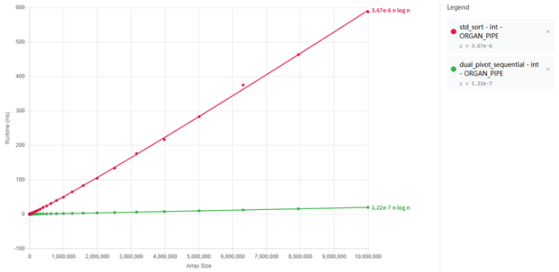


Figure 6: Organ-pipe

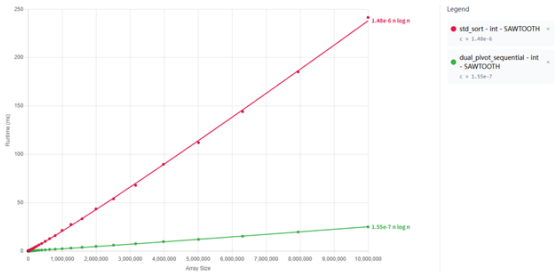


Figure 7: Sawtooth

One forward scan detects the runs; one merge pass fuses them.
Full sort collapses to linearithmic work in r = #runs, often $r \ll \log n$.

4.5 / 4.6 — Pivots From a 9-Comparator Network

Pick 5 samples anchored to the two ends:

```
step = (size / 8) * 3 + 3   // ~ 3/8 size
e1   = low  + step          // ~ 3/8 point
e5   = high - step          // ~ 5/8 point
e3   = (e1 + e5) / 2        // middle
e2   = (e1 + e3) / 2
e4   = (e3 + e5) / 2
```

Samples sit inside the central quarter, giving more balanced pivots.

Sort with hardcoded network — `sort5_network` = 9 comparators, branch-free.

Then branch on ordering:

```
if a[e1] < a[e2] < a[e3]
         < a[e4] < a[e5]:
    # strictly ordered
    DUAL-PIVOT  (4.6a)
```

4.6a — Dual-pivot partition [2]

```
[ < P1 ][ P1 <= x <= P2 ][ > P2 ]
 P1 = a[e1],  P2 = a[e5]
 backward scan + prefetch
```

4.6b — Dutch National Flag [5]

```
[ < P ][ = P ][ > P ]
 P = a[e3]
 middle region excluded from recursion
```

4.6 — Duplicates Route to DNF Automatically

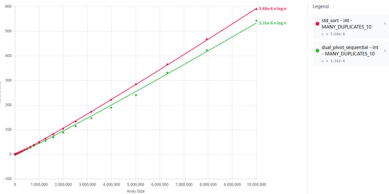


Figure 8: 10% unique

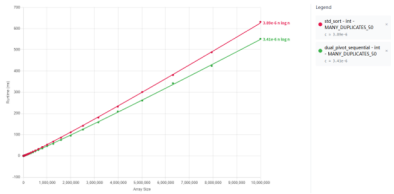


Figure 9: 50% unique

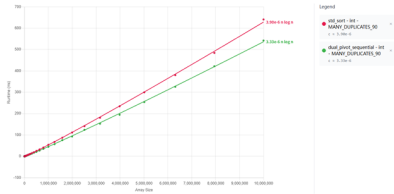


Figure 10: 90% unique

Absolute vs `std::sort` (`int`, $n = 10^7$): **1.09–1.18**$\times$ across the whole duplicate range — widest margin at *light* duplication, not heavy.

Why no huge win over our own random baseline? The input is still randomly permuted, so (i)~run-merging bails out — constant sub-sequences are length~1, (ii)~the 5-sample oracle only flags duplicates probabilistically, firing reliably once sub-arrays shrink enough that 5 random draws collide, (iii)~each DNF call absorbs only one pivot-value's band — typically 10–20% of the sub-problem.

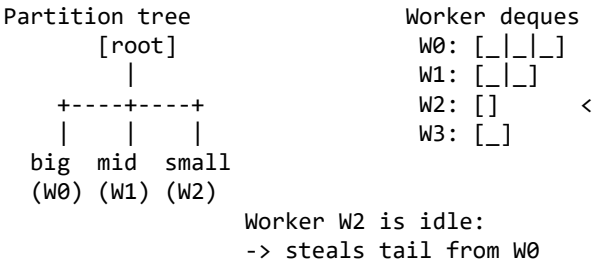
Why we still win against `std::sort`. libstdc++ is 2-way introsort with no DNF. Heavy duplicates luckily centre its pivots, so it stays competitive there; at moderate duplication its pivot quality degrades and our DNF keeps a steady edge.

Sequential Summary — One Table, Every Pattern

Pattern	Speedup	Responsible guard
Already sorted	near-instant	Early termination (Step 1)
`int8_t` / `int16_t` random	$\sim 8\times$ / $\sim 6\times$	Counting-sort dispatch (Step 2)
Random `int`	$\sim 1.0\times$ (parity)	Insertion-sort leaves (4.1 / 4.2)
Nearly-sorted	$\approx 0.74\times$ (slower)	`try_merge_runs` bails out (4.3)
Reverse-sorted	$\approx 10\times$**, near-linear**	`try_merge_runs` (4.3)
Many duplicates (any mix)	1.09–$1.18\times$	DNF partition (4.6b)

Every number above comes from the same hardware, same 10^7 elements, same methodology.

Parallel Path — Work-Stealing Thread Pool



Rules

- Push two larger sub-ranges to own deque; keep smallest (loop)
- Idle worker steals from another deque's tail (sticky-victim)
- `MIN_PARALLEL_SORT_SIZE` cutoff prevents micro-tasks

Start-up (few tasks, many idle workers)
At launch only the root partition exists — 1 task, N-1 idle workers. Idle workers consult a global "done" flag; while

Scaling — 10^7 random int

T	Time (ms)	Speedup	Eff.
1	513.4	1.00×	100 %
2	394.7	1.30×	65 %
4	159.8	3.21×	80 %
8	**108.7**	**4.72×**	59 %
16	244.9	2.10×	13 %

Peak at 8T; 16T regresses (see VTune slide for root cause).

Scaling Chart + the 8T Peak

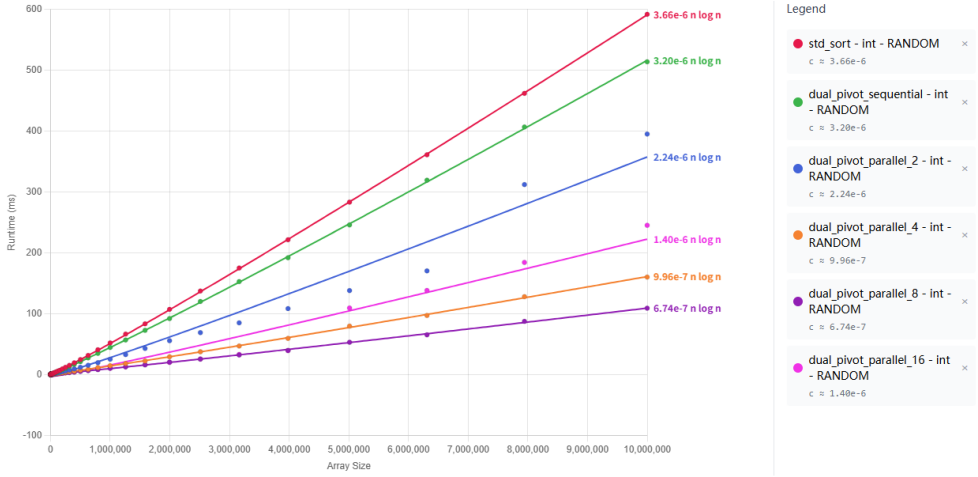


Figure 11: 10^7 random int — std::sort vs dpqs at 1 / 2 / 4 / 8 / 16 threads

Near-linear through 4T $\rightarrow$ sweet spot at **8T (4.72×)** $\rightarrow$ **regression at 16T**.

VTune Root-Cause — Four Stacked Ceilings (8T vs 16T)

Metric	8T	16T	Δ	Diagnosis
L3-Bound (% ticks)	14.5%	**25.1%**	**+10.6 pp**	**(1)** Working set overflows 20 MB L3
Machine Clears (% slots)	1.8%	**7.5%**	**×4.2**	**(3)** SMT memory-order nukes
L2-Bound (% ticks)	0.0%	1.0%	+1.0 pp	**(3)** SMT siblings share L2
L1-Bound (% ticks)	13.8%	14.7%	+0.9 pp	**(3)** SMT siblings share L1D
Slow-Pause spin-wait	0.0%	0.8%	+0.8 pp	**(4)** Mutex/steal contention
E-core clockticks	10.9G	**29.8G**	**×2.7**	**(2)** Work spilled to slower cores
E-core CPI	1.37	1.37	—	E-core ≈20% slower / instr than P-core (1.14)
P-core CPI	1.14	**1.55**	**+36%**	Net: every instr pays more cycles
DRAM-Bound	0.6%	0.5%	≈0	**Bandwidth is NOT the bottleneck**

Four stacked effects: **(1)** L3 latency ; $\gg$; **(2)** hybrid-core dilution ; $>$; **(3)** SMT cache sharing ; $>$; **(4)** mutex contention.

The silicon is the ceiling — not our algorithm, not our thread pool.

Engineering — "Using C++ Well"

- **Header-only, zero deps** — one `#include`, any C++17 build
- **Fully generic templates** — `T`, iterator category, `Compare` all type-parameters
- **STL-compatible iterator adapter**
  - contiguous → raw-pointer fast path (zero-cost)
  - non-contiguous → temporary buffer
- **`if constexpr` compile-time dispatch** — counting / float / dual-pivot decided at compile time
- **Hardware hints** — `DPQS_PREFETCH_READ` wraps `__builtin_prefetch` on the partition hot loop

Modular headers

```
include/
 dual_pivot_quicksort.hpp    (façade)
 dpqs/
   sequential_sorters.hpp
   run_merger.hpp
   counting_sort.hpp
   thread_pool.hpp
   utils.hpp
```

Limitations, Reflection & Conclusion

Honest Limitations

- Not stable — like `std::sort`, unlike `std::stable_sort`
- Move-only types — run-merger needs a copy path
- Nearly-sorted regression at some sizes — threshold tuning WIP

Reflection

> The biggest lesson wasn't the algorithm — it was that the real engineering is the **7 800-configuration harness + multi-seed methodology** that makes every tuning decision defensible.

Delivered

- Header-only, STL-compatible DPQS for C++17
- Adaptive: counting sort · run-merger · DNF · introsort fallback
- Work-stealing parallel — **4.72**× peak
- VTune-validated root-cause for the scaling curve

Section 7 — Q & A

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2. Research Gap & Methodology
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4. Parallel Path
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6. Engineering, Limitations, Conclusion
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Questions?

Thank you for your attention.

References

[1] C. A. R. Hoare, "Quicksort," *The Computer Journal*, vol. 5, no. 1, pp. 10–16, 1962.

[2] V. Yaroslavskiy, "Dual-pivot quicksort algorithm," Research report, 2009. [Online]. Available: http://codeblab.com/wp-content/uploads/2009/09/DualPivotQuicksort.pdf

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Backup — `sort5_network` (9 Optimal Comparators)

```
// e1 e2 e3 e4 e5  (indices into the array)
 1:  (e1,e2)   if a[e1]>a[e2] swap
 2:  (e4,e5)   if a[e4]>a[e5] swap
 3:  (e3,e5)   if a[e3]>a[e5] swap
 4:  (e3,e4)   if a[e3]>a[e4] swap
 5:  (e2,e5)   if a[e2]>a[e5] swap
 6:  (e1,e4)   if a[e1]>a[e4] swap
 7:  (e1,e3)   if a[e1]>a[e3] swap
 8:  (e2,e4)   if a[e2]>a[e4] swap
 9:  (e2,e3)   if a[e2]>a[e3] swap
```

Properties

- Proven optimal: 9 is the minimum comparator count to sort 5 elements
- Data-dependency depth = 6 → pipelines well on modern OoO CPUs
- Branch-free conditional swap via `std::min` / `std::max` (`cmov`)

Backup — Multi-Seed Protocol (Why Median-of-Medians?)

```
for seed in {42, 43, ..., 51}:                  # 10 seeds
    data = generate_random(size, seed)
    for w in 1..3:  sort(copy_of(data))         # warmup
    times = []
    for i in 1..10: times.push(time(sort(copy_of(data))))
    per_seed_median = median(times)             # absorbs run jitter
medians = [...]                                 # 10 per-seed medians
representative = median(medians)                # absorbs one unlucky seed
```

- **Mean across seeds** → skewed by one bad permutation
- **Min across seeds** → rewards one lucky permutation
- **Median-of-medians** → robust to both; aligns with current HPC benchmarking practice

Backup — Full Hardware Spec (from CPU-Z)

CPU — Intel Core i5-12600KF

Field	Value
Generation	12th Gen, Alder Lake
Socket	LGA1700
Process	10 nm
Cores / threads	10 (6P + 4E) / **16**
Max TDP	125 W
Core voltage	1.172 V
Core #0 speed	$□$4489 MHz
Multiplier	$×$45.0 (range 48.0–49.0)
Bus speed	99.76 MHz

Caches

Level	Size
L1-D	6 × 48 KB + 4 × 32 KB
L1-I	6 × 32 KB + 4 × 64 KB
L2	6 × 1.25 MB + 2 × 2 MB

Memory — 32 GB DDR5

Field	Value
Type	DDR5
Size	32 GB
Channels	4 × 32-bit (dual-channel)
DRAM frequency	2992.7 MHz (eff. $□$5985 MT/s)
Memory ctrl freq	1496.3 MHz
LLC / Ring	3591.2 MHz

Primary Timings

Param	Clocks
CAS Latency (CL)	36
tRCD	38
tRP	38
tRAS	80
tRC	118
Command Rate	2T